AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1350	**Classification:**	Critical / Chiral & Enantiomeric Purity (CQA)
Product / API Name:	Rosuvastatin Calcium API (USP/EP)	**Batch / Lot Number:**	RSV-2026-077
Manufacturing Stage:	Stage 2: Asymmetric Catalytic Hydrogenation	**Equipment Tag:**	High-Pressure Autoclave HYD-01 / Chiral HPLC
Date of Occurrence:	2026-10-14 09:00 UTC	**Date of Discovery:**	2026-10-15 16:15 UTC
Material / Lot ID:	Tetrahydrofuran Anhydrous RAW-THF-110	**Reporting Unit:**	QC Chiral Separations Laboratory

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Enantiomeric Purity (3R,5S-Enantiomer)	**NLT 99.5% ee**	**98.1% ee [OOS]**	Chiralpak AD-H Chiral HPLC
Unwanted (3S,5R)-Diastereomer	**NMT 0.20%**	**1.42% [OOS]**	Chiral HPLC UV 242nm
Assay (Rosuvastatin Calcium)	98.0% - 102.0%	98.8%	HPLC
Total Other Related Substances	NMT 0.50%	0.31%	Reverse Phase HPLC

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

Chiral HPLC analysis of intermediate Rosuvastatin ester batch RSV-2026-077 indicated enantiomeric purity dropped to 98.1% ee with 1.42% of the enantiomeric impurity. Investigation revealed raw reagent tetrahydrofuran lot RAW-THF-110 contained 45 ppm of dimethyl sulfide stabilizer residue that poisoned the chiral ruthenium Ru-(S)-BINAP catalyst, lowering stereoselective facial coordination during ketone reduction.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. Entire reaction mass (1,200 L) held in tank TK-HYD-01 under nitrogen blanket.
2. Raw material vendor quarantine placed on THF lot RAW-THF-110.
3. Sulfur trace screening added to raw solvent release testing protocol.

Prepared by:	Dr. V. Kulkarni (Chiral QC Lead)	2026-10-15
Verified by:	A. Dubois (Chemical Synthesis)	2026-10-15
QA Approval:	Dr. S. Thornton (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**